

DEVA ANAND

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Education

MS, Computer Science, University of Massachusetts Amherst 2025–2027

Advanced NLP, Advanced Machine Learning, Optimization in CS, Systems for Data Science, Performance Engineering

BE, Computer Science and Engineering, Vels Institute of Science, Technology and Advanced Studies 2019–2023

Professional Experience

Culture & Morality Lab, UMass Amherst, Lab Assistant | *Amherst, MA* Jul 2026 – Present

- Sole engineer building and maintaining the lab's internal text-analysis platform (Python, **FastAPI**, React) as a self-serve tool for researchers, owning architecture, the production design doc, and all technical decisions end to end.
- Built a YAML model registry, a versioned construct library (**99 constructs from 38 questionnaires**), and a structured, machine-readable data-quality warning system; grew the backend to **40 tests** in CI and documented rollout decisions in a maintained design doc.

Wysa, Backend Engineer | *Bengaluru, India* Jul 2023 – Jul 2025

- Engineered a multi-tenant NHS eTriage backend (Node.js, MongoDB) for **8+ UK clinical clients**, processing **10,000+ monthly submissions** via Mayden iaptus REST integrations; contributed to **\$340K+** in revenue.
- Diagnosed and resolved production issues across tenants, monitoring logs during releases and client integrations and shipping targeted fixes for tenant-specific incidents to keep the multi-tenant service stable.
- Served as primary technical point of contact for the eTriage backend, communicating resolutions to stakeholders and running structured knowledge transfer that brought a new engineer to on-call readiness in ~6 weeks; designed authentication and audit logging for user access.
- Built an internal full-stack data-annotation tool (React, Node.js, MongoDB) used daily by the AI team, replacing a manual spreadsheet workflow and cutting repetitive work by **100+ hours/month** while lifting labeling throughput **60%**.

Freelance / Open Source, Full-Stack Engineer | *Remote* Jul 2022 – Aug 2023

- Delivered and maintained production web platforms for international clients, including an **inventory management system** for a US client and a gym management platform (MERN, live **3+ years**) with member, attendance, and payment record tracking, git-webhook auto-deploy, and cron-based automation.

Projects

Sonare – Offline Sign / Speech Desktop App (GitHub) React, Electron, FastAPI, MediaPipe, whisper.cpp

- Built a cross-platform desktop application (Windows/macOS/Linux via Electron + React and a FastAPI backend) integrating camera, microphone, and on-device models (MediaPipe, whisper.cpp) into an offline pipeline with no external network dependencies.

Spark ETL Performance Optimization (GitHub) PySpark, Spark UI, SQL, NYC TLC Dataset

- Diagnosed and optimized a Spark/PySpark ETL pipeline over a **~1.4B-row** dataset, using Spark UI stage profiling to localize bottlenecks and cutting end-to-end runtime **~25%** (17–19 min to 14.3 min) via broadcast joins, Adaptive Query Execution, and skew-join handling.

AgenticSearch – Provenance-First Entity Discovery (GitHub | Demo) Python, FastAPI, OpenAI / Groq, SQLite

- Built a Python/FastAPI service that turns open-ended queries into structured entity tables with deterministic extraction and **cell-level provenance**; hardened with **150+ automated tests** and provider fallback for reliable operation.

Publications

- "Alzheimer's Disease Classification using Transfer Learning," IEEE CONIT 2023, first author, deep transfer learning on neuroimaging data. (Paper)

Skills

Languages: Python, Bash, SQL, JavaScript, TypeScript, Java, C++

Systems & OS: Linux, macOS, Windows (cross-platform Electron), Docker, Git, GitHub Actions

Troubleshooting: production debugging, log analysis, incident resolution, performance profiling, integration testing

Web & Data: React, Next.js, Node.js, FastAPI, REST APIs, PostgreSQL, MongoDB, SQLite

Cloud & DevOps: AWS, GCP, Azure, CI/CD, MongoDB Atlas

Open Source & Community

- Selected for **GirlScript Summer of Code 2023**; contributed to Linkfree, Freehit, and ProjectsHut. - Tutored **30+ underprivileged students** in programming at Sayur, a non-profit in Tamil Nadu.